2024 International Wireless Communications and Mobile Computing (IWCMC) | 979-8-3503-6126-1/24/$31.00 ©2024 IEEE | DOI: 10.1109/IWCMC61514.2024.10592416

Advanced Deep Learning Framework for Improved Wildfire Detection and Aerosol Identification Using Active Satellite Imagery

Srija Venkata Sai Ravali Kothapalli
School of Computing
National College of Ireland
Dublin, Ireland
srijaravalikothapalli@gmail.com

Cristina Hava Muntean
School of Computing
National College of Ireland
Dublin, Ireland
cristina.muntean@ncirl.ie

Abid Yaqoob
School of Electronic Engineering
Dublin City University
Dublin, Ireland
abid.yaqoob@dcu.ie

***Abstract*— Wildfires rank among the most prevalent natural disasters globally and have emerged as a significant factor in climate change over the past decade. Early detection of wildfires and smoke plumes through satellite imagery is crucial since they are not easily extinguishable which may lead to catastrophic consequences for both wildlife and forest ecosystems. Classic deep-learning models for wildfire and aerosol identification have shown significant progress, but high false-positive rates remain a key limitation. This paper proposes a custom-designed Convolutional Neural Network (CNN) model that aims to improve the identification of wildfires and aerosols, leveraging satellite imagery categorized into cloud, dust, haze, land, seaside, and smoke. Moreover, we considered popular deep learning and transfer learning models, specifically EfficientNet, MobileNetV3, and Inception V3, to identify and distinguish smoke plumes. Hyper-parameter tuning has been performed to achieve more accurate results. The metrics evaluated in this research are accuracy, precision, recall, and f1-score. A comprehensive analysis was performed that aimed to identify the best transfer learning model and the model that closely aligns with the performance of the CNN built. This paper provides valuable insights into the potential use of transfer learning models and of the proposed custom-designed CNN model in early wildfire detection by identifying the smoke plumes and helping in reducing false fire alarms.**

Keywords—Wildfires, Satellite Imagery, Convolutional Neural Network, Image Classification, Transfer Learning Models

I. Introduction

Forests play the most crucial role in sustaining life on Earth and contribute significantly to the well-being of both the environment and human societies by providing various natural resources. They almost cover over 31% of the land area on the Earth. The escalation of wildfires, exacerbated by the effects of climate change, has become a critical global concern due to the cataclysmic impact on the forests and wildlife. The frequency of wildfires has been a major concern during the last decade. Wildfires may occur anywhere in the forests and impact climate change by affecting various sustainable natural resources, and wildlife. Climate change is responsible for exacerbating droughts, heatwaves all over the planet, and the flames emerging in many forests.

The year 2022 was the second-worst wildfire season in the European Union since 2000 when the Copernicus' European Forest Fire Information System (EFFIS) records began [1]. Spain was the most affected by wildfires producing around 1.3 million tons of carbon emissions [2] in just one month between June 2022 to July 2022 with a total affected area that is three-and-a-half more than in 2021. The fire events that occurred in Spain are one of the major examples of the destruction. Many such incidents are affecting the planet by causing various ecological impacts on biodiversity hotspots, air quality, natural resources, rainfall patterns, and wildlife. All of these indicate that there is a need for advanced technologies to be used to detect wildfires in the early stage by identifying the smoke plumes through satellite imagery and predicting the actual locations without misleading by false fire alarms. Various studies and experiments have been conducted to address these issues. Artificial Intelligence (AI) and Machine Learning (ML) techniques have been proposed to be used for detecting wildfires [3] as early as possible. The adaptive transmission of wildfire data can also be of particular interest where the goal is to visualize the streaming content in higher quality [16], [17] [18].

In this paper, we propose an ***advanced deep-learning framework for improved wildfire detection and aerosol identification using active satellite imagery***. To address the challenges posed by limited wildfire and smoke datasets, we utilize the University of Science and Technology of China (USTC) smoke dataset captured using MODIS (Moderate Resolution Imaging Spectroradiometer) instrument [4], where the image data is collected from different regions related to six different classes such as cloud, dust, haze, land, seaside, and smoke. Our core contributions focus on refining smoke plume detection and combating false alarms. The main contributions of this work are as follows:

- Development of a custom CNN model, fine-tuned through hyperparameter adjustments, to accurately identify smoke, haze, dust, and similar atmospheric phenomena.
- Evaluation of popular transfer learning models (EfficientNet, MobileNetV3, and InceptionV3) for smoke plume identification and distinction.
- Comprehensive experimental analysis highlighting the benefits of both transfer learning and the custom CNN model for early wildfire detection, with results validated using accuracy, precision, recall, and f1-score metrics.

The rest of the paper is structured as follows: Section II reviews the latest research works in the area. Section III presents state-of-art models, and the newly built CNN architecture, and Section IV presents the implementation and experimental evaluation of the proposed models and provides the steps involved in the testing process of each model followed by their corresponding results. Finally, the conclusion and future work are presented in Section V.

II. Related Works

Deep learning methods have been the major problem-solving methods used for wildfire detection procedures. Convolutional neural networks have been used frequently in various deep-learning approaches for forest fire detection. CNN for automated wildfire is proposed by Khryashchev et al. [5] where two datasets having satellite RGB images with different spatial resolutions (1457 images) and high-

979-8-3503-6126-1/24/$31.00 ©2024 IEEE

Authorized licensed use limited to: University of Missouri Libraries. Downloaded on March 03,2026 at 23:52:00 UTC from IEEE Xplore. Restrictions apply.

resolution images (393 images). An Adaptive moment estimation algorithm has been used in this paper for training process optimization.

Classification of images using Artificial Neural Networks (ANNs) proposed by Nabi et al. [6] is different from existing studies in terms of training several different architectures and classifying the images related to forest fires. Two CNNs have been created for fire detection in terms of classification and segmentation. CNN's and hyperspectral data analysis has been performed by Thangavel et al. [7] for accelerating the hardware. The researchers considered three distinct hardware accelerators like the Intel Movidius Myriad 2, the Nvidia Jetson Nano, and the Nvidia Jetson TX2 to show that the onboard application, which is practicable for upcoming or future space missions, helps in improving the services as well as in establishing and managing the space to ground data flow by providing the data in real-time. Deep CNN "MultiScale-Net" has been proposed by Rostami et al. [8] for AFD (Active Fire Detection) where the Landsat-8 imagery datasets have been used at the pixel level. To improve the performance, AFI (Active Fire Index) is added to the network as their inputs have shortwave infrared (SWIR2, SWIR1), and blue bands were also aided. IoU scores were also obtained along with highest precision and sensitivity and f1-score. Pereira et al. [9] also used the same Landsat-8 satellite imagery dataset where 146,214 image patches have been extracted and split into two parts containing the spectral images with 10-band.

Reis et al. [10] proposed a fire luminosity airborne-based ML evaluation dataset having images of forest fires. The images were obtained using an unmanned aerial vehicle. Deep-learning algorithms such as ResNet50V2, DenseNet121, Inception V3, and VGG-19 have been used. Images using AI-based computer vision techniques have been introduced by Sathishkumar et al. [11]. Transfer learning models such as VGG16, Xception, and Inception V3 were used. Sousa et al. [12] presented a transfer learning model approach based on Inception V3. The research involved combining data augmentation techniques which have been tested under the tenfold cross-validation scheme. Almost 35 real fire events have been covered by featuring images from open-source dataset. Similarly, Inception V3 transfer learning model has been utilized by Vani et al. [3]. The proposed model is CNN-based Inception V3 model which can extract the images automatically by categorizing the images into two sets such as fire images and non-fire images.

The research presented in this paper focuses on exploring the most important algorithms helpful in predicting the smoke plumes easily by analyzing their performance. We have used three transfer-learning models such as EfficientNet, MobileNetV3 and Inception V3. These models are implemented through TensorFlow Framework, we have also built a new CNN model by including more layers, building blocks in the architecture with normalization and drop-out layers. The major difference in this research paper compared to other papers is addressing the aerosol classes by performing the analysis on six different classes additionally with hyper-parameter tuning where the models are fine-tuned by reducing the processing times for the best and accurate results.

III. Proposed Research Methodology and Architecture

The research work presented in this paper aimed on building a new CNN model from scratch as CNNs are the best models for dealing with complex images. Also, the proposed model is compared with three different cutting-edge transfer learning models on a dataset with satellite imagery. Fig. 1 illustrates the Implementation pipeline used in this paper where the architecture of the proposed research methodology follows different steps such as Data collection, Pre-processing and transformation, Design specifications, Experimental implementation and Evaluation, and then finally concluding the best model obtained in this research.

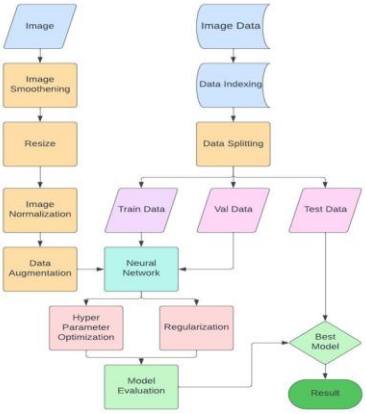

Fig. 1. Implementation Pipeline

Data Collection: USTC-Smoke dataset was used in this research, where the data was extracted from the MODIS sensors that have been installed on the Tera and Aqua satellites that are used extensively for wildfire detection. These satellites overpass 4 times a day and revisit the locations on the Earth every 1-2 days. The dataset has 6225 RBG images relevant to wildfires, classified into six categories, such as Smoke, Dust, Haze, Land, Cloud, and Seaside. The dataset has been obtained from the Kaggle[1].

Data Preparation: Due to some comparable spectral and feature textures Cloud, Dust and Haze categories are identified as one of the most similar classes in the dataset. Where Land and Water image categories have been included to differentiate various terrains. Hence the extracted dataset is geometrically corrected, where channels 1, 3, 4 are employed to obtain the RGB images from the satellite [13]. The data has been gathered from events that occurred throughout the twenty years where the MODIS/Terra and Aqua Thermal anomalies are helpful in identifying fire-related smoke, hence all these images are used for training. More details on how images were manually acquired and labelled are provided in [14]. Sample images from each category are shown in Fig. 2.

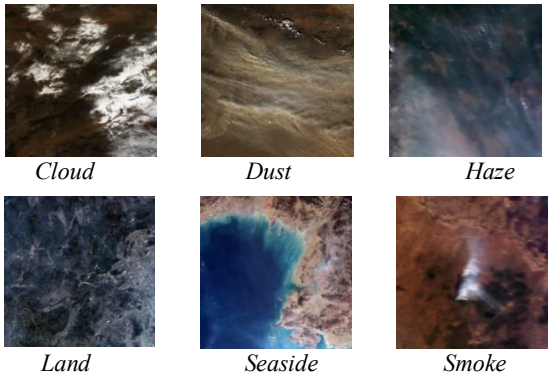

Fig. 2. Six Classes of Aerosols in the Satellite imagery Dataset

Data Pre-Processing and Transformation: It is critical to integrate the image processing steps and to consider the models capability of distinguishing even the smallest patterns or details. Convolutional neural networks are the best models to deal with such data having complex images. So, we have implemented the CNN from scratch and compared it to a number of advanced transfer learning models that were

[1] https://www.kaggle.com/datasets/horton1314/ustssmoker

Authorized licensed use limited to: University of Missouri Libraries. Downloaded on March 03,2026 at 23:52:00 UTC from IEEE Xplore. Restrictions apply.

applied on the dataset. Image smoothening has also been performed to eliminate or remove the disturbances that occur in such high-resolution images such as noise, clutter, and sharpness from an image. This process results in giving a smooth and blended effect, the edges having the disturbances are identified and the unnecessary noise is removed. Below Fig. 3 shows the sample images outputs after smoothening and removing unwanted noise.

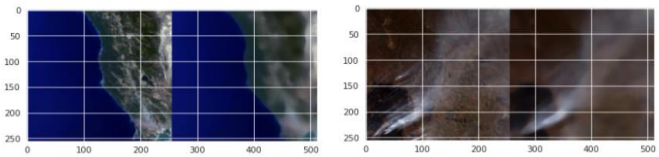

Fig. 3. Image-Smoothening Process

Original images have been resized to smaller image sizes to enable the model to perform faster and better. Normalization is performed to converge the model with its best optimal parameters. Image augmentation is also performed for converting the images into various forms. Edge detection and image recognition factors can be performed using the linear transformations techniques [15]. Below Fig. 4 shows the image after the application of the linear transformation where smoke and cloud can be distinguished. The cloud is more distributed, and the smoke starts at a point where it has some shape of a tail.

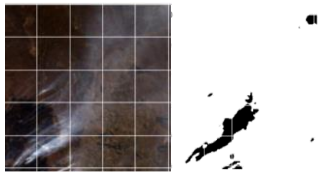

Fig. 4. Linear-Transformation

Convolutional Neural Network (CNN) Model Design: To build a sequential Convolutional Neural Network from scratch mechanisms such as continuous spatial and channel attention are required. Here, all input channels are integrated as one matrix for spatial attention. This matrix has been generated using element-wise multiplication. Where the channel attention is responsible for the construction of a single scalar from every pixel to pay respective attention to each channel. The proposed CNN model has been built to transfer the data or corresponding information from each layer to the final layer. As we progress around the network, this results in conveying better data in terms of providing information with less trainable parameters. To prevent overfitting, batch normalization and dropout layers are inserted. This proposed CNN model was designed to be an effective and efficient one considering its output parameters and performance at each stage with the resources that are available.

TABLE I
CLASSIFICATION REPORT FOR INCEPTIONV3 MODEL

Layer Name	Output Shape
Zero Padding + Conv 2 * 2	120*120*3
Batch Normalisation + ReLu + Zero padding + conv2d (Conv2D) + Dropout + Block Concatenate	120*120*32
Batch Normalisation + ReLu + Zero padding + Conv 2 * 2 + Dropout + Block Concatenate	120*120*64
Batch Normalisation + ReLu + Conv2D+ Dropout	120*120*60
Average Pooling	60*60*60
Batch Normalisation + ReLu + Zero padding + Conv 2 * 2+Dropout + Block Concatenate	60*60*108
Batch Normalisation + ReLu + Zero padding + conv2d (Conv2D) + Dropout + Block Concatenate	60*60*172
Batch Normalisation + ReLu + Conv2D+ dropout	60*60*30
Average Pooling	30*30*30
Batch Normalisation + ReLu + Zero padding + Conv 2 * 2 + Dropout + Block Concatenate	30*30*110
Batch Normalisation + ReLu + Zero padding + Conv 2 * 2 + Dropout + Block Concatenate	30*30*206
Batch Normalisation + ReLu + Conv2D+ dropout	30*30*15
Average Pooling	15*15
Global Average Pooling	None,15

Table I details about the design of proposed CNN model. Aiding the Keras Tuner, the initial parameters required, and performed hyper-parameter tuning to train the CNN with different output parameters such as learning rates and drop-out rates to obtain the best optimal values for the dataset.

1. Modelling Techniques

1.1 Transfer Learning Models

Transfer-Learning models are the pre-trained models used to resolve various problems related to classification. Currently, these models are very popular through the deep learning approaches as they can train deep neural networks and can be used for training the new models. By loading on and evaluating these models, we can obtain the best model where large amounts of standard data having very high and good computational power shows the trade between the performance and speed. In this paper, transfer learning models has been considered for research based on their performance and speed are as follows:

- **EfficientNet:** This model's application with the benchmark datasets has significant results compared to other transfer learning models. This architecture uses the Compound coefficient technique to ensure the width and depth are scaled uniformly. It is also proved that their efficiency to reduce various parameters and corresponding floating-point operations varies for each second.
- **MobileNet:** MobileNetV3Large has been used in the proposed model design where the depth-wise separable convolution has been introduced which is a new kind of convolution layer. These convolutions with the depth-wise separable layers are more efficient compared to the standard 2D convolution operations. MobileNet models are significantly faster and takes very less time by making the size small which makes them easily accessible from the mobile app applications.
- **InceptionV3:** The Inception-v3 model is based on the database related to ImageNet, where the main theme behind this model is to utilize all the corresponding operations in a particular single layer. This implies that instead of separate layers for max pooling and convolution operations, we can have two operations in each layer. This makes the model different from other models as well.

1.2 AUC-ROC CURVE

AUC-ROC curves are used in classification-related problems to measure performance, Here AUC (Area under the curve) is known for the measure of separability and the ROC (Receiving Operating Characteristics) stands for Curve Probability. We have used the AUC-ROC curve to understand how better the models are differentiating various classes of aerosols. We obtained the plots related to each class of images regarding the model's performances and the plots are between True Positive Rate and False Positive Rate. The analysis using the AUC-ROC curve gives us a better understanding of all the models investigated in this research. Hence all these models are implemented with this AUC-ROC Curve feature in the end results.

Authorized licensed use limited to: University of Missouri Libraries. Downloaded on March 03,2026 at 23:52:00 UTC from IEEE Xplore. Restrictions apply.

IV. Experimental Evaluations

We have performed a thorough examination of all models investigated in this research such as Transfer-Learning models and the newly built CNN model. Various evaluation metrics have been considered in terms of confusion matrix which involves accuracy, precision, recall, and f1-score. Here we have also presented the respective AUC-ROC Curves for all the trained models. We conducted our experiments using the Google Colab platform, leveraging a Python 3 environment on Google Compute Engine, specifically utilizing GPU resources. The hardware accelerator selected for these experiments was a T4-GPU, configured with a High-RAM profile. Satellite imagery dataset has been mounted from Google Drive to Google Collaboratory. There are six different folders related to six different classes or categories of aerosols such as Cloud, Dust, Haze, Land, Seaside, and Smoke. The dataset contains 6225 RGB images which are divided into three subsets such as 85.5% of training data, 4.5% of validation data, and 10% of test data. The Python programming language is used for performing the pre-processing which includes all the image transformations, smoothening, normalization, data indexing, and data augmentation steps. Also, Hyper-parameter tuning has been carried out for the models to obtain the best output parameters. Keras and TensorFlow libraries have been used accordingly. In terms of learning rate values, we have considered various values between 0.01 and 0.0001 and tested them for the hyperparameter selection. The model has been fine-tuned for each dropout rate and growth rate. Ideally, high values of the learning rates led to overfitting and low values have led to model accuracy becoming stationary. Therefore, we have set the learning rate for 0.00075 to avoid this overfitting issue. ADAM optimizer is used for optimizing or to converge the respective models. Initially, the same parameters as optimization strategy and cost function have been followed. We have performed around 20 epochs to get better and consistent accuracy for CNN and 25 epochs each for all the transfer learning models. Initially, each model was involved in training on the trained data to make the model understand and learn the data better, Validation on validation data for measuring the model performance. In this step, we decided on the final parameters while validating the data. The training parameters which gained the highest validation accuracy for the model are taken as the best optimal parameters and considered for an efficient model.

1. Inception V3 Model Evaluation

The Inception V3 model has achieved very poor accuracy compared to other models, it is unable to distinguish between the aerosols accurately such as haze and smoke, Edge detection has been poor. The f1-score for the model in each category helps us to understand that model has failed to classify between Haze, Land, and Smoke.

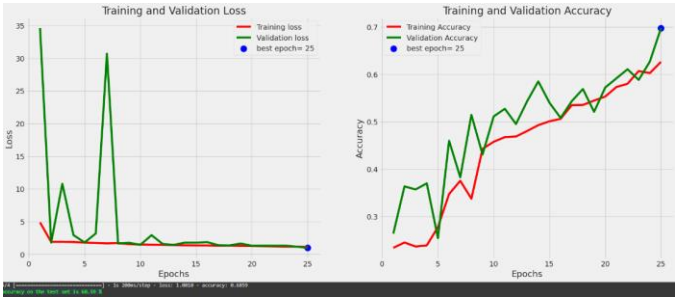

Fig. 5. InceptionV3 Model Validation and Testing accuracies Plot

The above Fig. 5 shows the different epoch values tested concerning the model for the smoke dataset used. The model achieves the best accuracy at 25th epoch. The loss for this model was settled at 1.0010. Coming to the test set accuracy, InceptionV3 model achieved around 68.59%. Below Table II, the classification report for the Inception V3 model with respect to all the aerosol classes can be observed with respective evaluation metrics.

TABLE II
Classification Report for InceptionV3 Model

	Precision	Recall	F1-Score	Support
Cloud	0.81	0.79	0.80	58
Dust	0.61	0.82	0.70	57
Haze	0.58	0.46	0.51	48
Land	0.56	0.84	0.67	49
Seaside	1.00	0.82	0.90	51
Smoke	0.64	0.33	0.43	49
Accuracy				312
Macro Avg.	0.70	0.68	0.67	312
Weighted Avg.	0.70	0.69	0.68	312

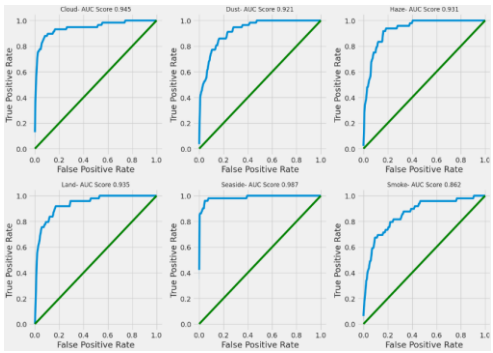

Fig. 6. AUC-ROC Curves for InceptionV3 Model

Fig. 6 illustrates the output obtained for the AUC-ROC curves in the InceptionV3 model. The more the area lies under the respective curve, the better the respective performance is measured in AUC-ROC curves. Unfortunately, Inception V3 model accuracy is low, particularly in the case of Dust, Haze, and Smoke.

2. MobileNetV3Large Model Evaluation

MobileNet is one of the best architectures that uses depth-wise separable convolution layer. It has outperformed at its best compared to the Inception V3 model by efficiently distinguishing all the aerosols comparatively. This model has distinguished all the aerosols accurately and was able to identify the Smoke class with the f1-score of 91%. This is the model which has fewer parameters with faster training rates that obtained the most accurate results on all the aerosol classes.

TABLE III
Classification Report for MobilenetV3 Model

	Precision	Recall	F1-Score	Support
Cloud	0.92	0.98	0.95	58
Dust	0.91	0.89	0.90	57
Haze	0.94	0.96	0.95	48
Land	0.94	0.94	0.94	49
Seaside	0.98	0.96	0.97	51
Smoke	0.93	0.88	0.91	49
Accuracy				312
Macro Avg.	0.94	0.94	0.94	312
Weighted Avg.	0.94	0.94	0.94	312

Authorized licensed use limited to: University of Missouri Libraries. Downloaded on March 03,2026 at 23:52:00 UTC from IEEE Xplore. Restrictions apply.

From Table III, the classification report for MobileNetV3Large model can be observed, the model has achieved accuracy of 94% by outperforming the InceptionV3 model. From the plot obtained between the training loss and validation loss for this model with 25 epochs the best output parameters were obtained at epoch =21. The second plot between model's training accuracy and validation accuracy, the best accuracy is obtained at epoch=19. Accuracy on the test set is around 93.59% with the loss settling at 0.4998.

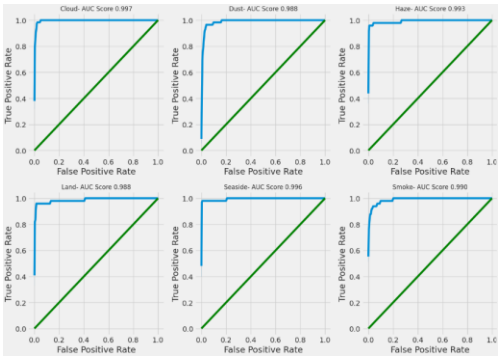

Fig. 7. AUC-ROC Curves for MobileNetV3Large Model

By observing the curves from Fig. 7, we can understand that the transfer learning model has performed good by predicting respective aerosol classes accurately as we can see that the AUC-ROC curves related to each class have more area under the curve.

3. EfficientNet Model Evaluation

This model is one of the best and the most efficient models, which has the efficient architectures till date as it uses advanced mathematical techniques such as compound coefficient. In terms of performance, EfficientNet has outperformed all the other models proposed in this paper and the accuracy of both the training and testing datasets is high.

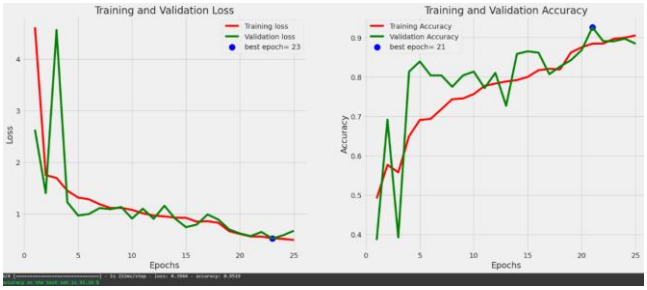

Fig. 8. EfficientNet Model Validation and Testing accuracies Plot

From the plots in the above Fig. 8, we have observed that the test set accuracy is 95.19% with the loss setting at 0.3944. The major difference between the MobileNetV3 and EfficientNet is that the former was not able to identify the difference between Smoke and Haze clearly and few errors were seen in confusion matrix results.

TABLE IV
CLASSIFICATION REPORT FOR EFFICIENTNET MODEL

	Precision	Recall	F1-Score	Support
Cloud	0.97	0.98	0.97	58
Dust	0.90	0.98	0.94	57
Haze	0.92	0.94	0.93	48
Land	0.96	0.90	0.93	49
Seaside	1.00	0.96	0.98	51
Smoke	0.98	0.94	0.96	49
Accuracy				312
Macro Avg.	0.95	0.95	0.95	312
Weighted Avg.	0.95	0.95	0.95	312

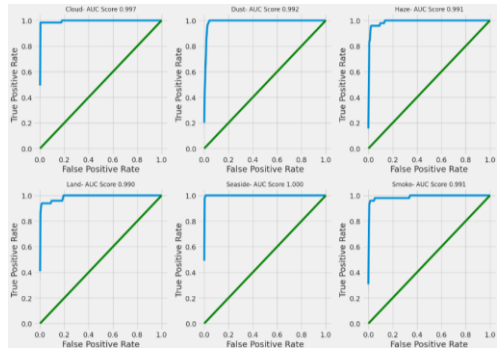

Fig. 9. AUC-ROC Curves for EfficientNet Model

In the above Fig. 9, the area under the curves is maximum for almost all the aerosol classes which is almost similar in the MobileNetV3 model AUC-ROC curves.

4. New Convolutional Neural Network Model

CNN has been built from the scratch to achieve better results in predicting the aerosol classes like other transfer learning models which are pre-trained. Similarly, we have used heavy GPUs from google colab to obtain the most accurate results with respect to our new CNN model proposed.

Below are the output parameters for this new CNN model in Fig. 10 where the training and the validation results are obtained and at epoch=16 the model achieved better accuracy with the loss settling around 0.80.

Fig. 10. New CNN Model Validation and Testing accuracies Plot

TABLE V
CLASSIFICATION REPORT FOR NEW CNN MODEL

	Precision	Recall	F1-Score	Support
Cloud	0.87	0.81	0.84	58
Dust	0.83	0.79	0.81	57
Haze	0.74	0.67	0.70	48
Land	0.62	0.84	0.71	49
Seaside	0.95	0.75	0.84	51
Smoke	0.69	0.78	0.73	49
Accuracy				312
Macro Avg.	0.79	0.77	0.77	312
Weighted Avg.	0.79	0.77	0.78	312

The above Table V details that the newly built CNN model has almost predicted all the classes almost equally by classifying all the aerosol class images from the dataset. Here in the above classification report, the f1-score of the model says that land and haze classes are predicted less compared to the other classes whereas cloud, dust, seaside, and smoke are predicted almost accurately.

Authorized licensed use limited to: University of Missouri Libraries. Downloaded on March 03,2026 at 23:52:00 UTC from IEEE Xplore. Restrictions apply.

Below Fig. 11 illustrates the AUC-ROC curve output where we the best performance has been obtained for our newly built CNN model compared to the AUC-ROC Curve results of InceptionV3 model. Here, the model has decently predicted all the aerosol classes successfully by having the AUC-ROC score obtained between the True Positive Rate and the False Positive Rate in each class is more than 0.90 where haze and land has the lower score with less area under the curve compared to other aerosol class curves like cloud, dust, seaside, and smoke.

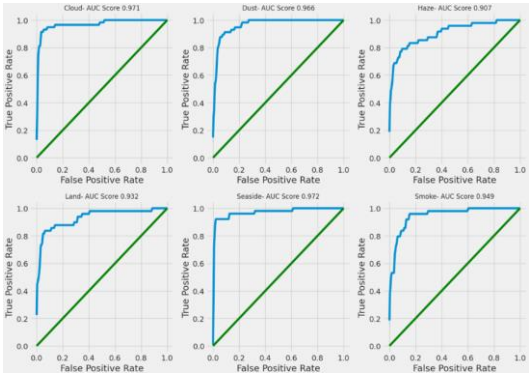

Fig. 11. AUC-ROC Curves for the New CNN Model

TABLE VI
MODEL COMPARISON

Model	Accuracy	Precision	Recall	F1-Score
InceptionV3	0.69	0.70	0.69	0.68
MobileNetV3Large	0.94	0.94	0.94	0.94
EfficientNet	**0.95**	**0.95**	**0.95**	**0.95**
New CNN	0.77	0.79	0.77	0.77

Below Table VI shows the results for the 4 models analyzed. The results show that EfficientNet which is the latest Transfer-Learning model in the present deep-learning approaches has outperformed all the other proposed models by achieving an accuracy of 95% where MobileNetV3Large performed almost similar to EfficientNet model. The new CNN architecture proposed in this paper has performed well and outperforms InceptionV3, one of the most powerful transfer learning models we have chosen to research.

A combination of the various batch sizes, learning rates, and image sizes has been used to achieve high-quality classification results. We observed that reducing the image's size has decreased the model running time and increased the precision for both MobileNet model and the new CNN. But it was the opposite for the EfficientNet, the model with the larger images as its inputs performed better. In the new CNN, convoluted layers with the Max-Pooling method have led the model with low accuracy compared to the best performed transfer learning models. Using the Average Pooling method led to better results and satisfactory classification.

V. CONCLUSIONS AND FUTURE WORKS

This paper proposes a new deep CNN model for sensing smoke through satellite imagery and compares its performance with various state-of-the-art image processing models. The integration of multiple blocks connected enables for transfer of additional information within the network and aids in the discovery of complex patterns. This new CNN model requires a relatively small number of parameters; thus, it speeds up training. When the proposed model is compared to transfer learning models like Inception V3, our newly built CNN model performs satisfactorily. The extensive results indicate that EfficientNet model particularly stands out with a remarkable 95% accuracy rate, indicating its potential for real-time wildfire detection. In this paper, we have only pertained to specific spatial resolution. As future works, expanding the application to various spatial resolutions enhances the model's comprehension of satellite data. Leveraging multispectral data improves real-time smoke detection efficacy by capturing diverse frequencies and wavelengths. The insights derived in this paper not only inform strategic decision-making processes but also pave the way for the development of proactive wildfire management strategies. Ultimately, this research represents a significant step forward in the ongoing efforts to protect and preserve our natural environments in the face of escalating wildfire risks.

REFERENCES

[1] European Commission, Joint Research Centre, San-Miguel-Ayanz, J., Durrant, T., Boca, R. et al., Forest fires in Europe, Middle East and North Africa 2022, Publications Office of the European Union, 2023, https://data.europa.eu/doi/10.2760/348120

[2] "Wildfire Articles" Nasa.gov. https:// https://earthdata.nasa.gov/learn/articles/feature-articles/wildfire-articles

[3] K. Vani, "Deep learning-based forest fire classification and detection in satellite images," in 2019 11th International Conference on Advanced Computing (ICoAC), IEEE, 2019, pp. 61-65.

[4] A. Akbari Asanjan, M. Memarzadeh, P. A. Lott, E. Rieffel, and S. Grabbe, "Probabilistic Wildfire Segmentation Using Supervised Deep Generative Model from Satellite Imagery," Remote Sensing, vol. 15, no. 11, p. 2718, 2023.

[5] V. Khryashchev and R. Larionov, "Wildfire segmentation on satellite images using deep learning," in 2020 Moscow Workshop on Electronic and Networking Technologies (MWENT), IEEE, 2020, pp. 1-5.

[6] M. Nabi and I. Bordanov, "Research of Forest Fire Detection on Satellite Images," in 2023 International Conference on Industrial Engineering, Applications and Manufacturing (ICIEAM), IEEE, 2023, pp. 803-808.

[7] K. Thangavel, D. Spiller, R. Sabatini, S. Amici, S. T. Sasidharan, H. Fayek, and P. Marzocca, "Autonomous Satellite Wildfire Detection Using Hyperspectral Imagery and Neural Networks: A Case Study on Australian Wildfire," Remote Sensing, vol. 15, no. 3, p. 720, 2023.

[8] A. Rostami, R. Shah-Hosseini, S. Asgari, A. Zarei, M. Aghdami-Nia, and S. Homayouni, "Active fire detection from landsat-8 imagery using deep multiple kernel learning," Remote Sensing, vol. 14, no. 4, p. 992, 2022.

[9] G. H. de Almeida Pereira, A. M. Fusioka, B. T. Nassu, and R. Minetto, "Active fire detection in Landsat-8 imagery: A large-scale dataset and a deep-learning study," ISPRS Journal of Photogrammetry and Remote Sensing, vol. 178, pp. 171-186, 2021.

[10] H. C. Reis and V. Turk, "Detection of forest fire using deep convolutional neural networks with transfer learning approach," Applied Soft Computing, vol. 143, p. 110362, 2023.

[11] V.E. Sathishkumar, J. Cho, M. Subramanian, and O.S. Naren, "Forest fire and smoke detection using deep learning-based learning without forgetting," Fire Ecology, vol. 19, no. 1, pp. 1-17, 2023.

[12] M. J. Sousa, A. Moutinho, and M. Almeida, "Wildfire detection using transfer learning on augmented datasets," Expert Systems with Applications, vol. 142, p. 112975, 2020.

[13] Y. Xie, J.J. Qu, X. Xiong, X. Hao, N. Che, and W. Sommers, "Smoke plume detection in the eastern United States using MODIS," International Journal of Remote Sensing, vol. 28, no. 10, pp. 2367-2374, 2007.

[14] R. Ba, C. Chen, J. Yuan, W. Song, and S. Lo, "SmokeNet: Satellite smoke scene detection using convolutional neural network with spatial and channel-wise attention," Remote Sensing, vol. 11, no. 14, p. 1702, 2019.

[15] S. Miao, H. Lin, H. Gao, and L. Dong, "Strip smoke and cloud recognition in satellite image," in 2016 9th International Congress on Image and Signal Processing, BioMedical Engineering and Informatics (CISP-BMEI), IEEE, October 2016, pp. 303-307.

[16] A. Yaqoob and G. M. Muntean, "Fuzzy Logic-based Adaptive Multimedia Streaming for Internet of Vehicles," 2023 IEEE 97th Vehicular Technology Conference (VTC2023-Spring), Florence, Italy, 2023, pp. 1-6.

[17] A. Yaqoob, C. Muntean and G. -M. Muntean, "Flexible Tiles in Adaptive Viewing Window: Enabling Bandwidth-Efficient and Quality-Oriented 360°VR Video Streaming," 2022 IEEE International Symposium on Broadband Multimedia Systems and Broadcasting (BMSB), Bilbao, Spain, 2022, pp. 1-8.

[18] A. N. Moldovan and C. H. Muntean, "QoE-aware video resolution thresholds computation for adaptive multimedia", Proc. IEEE Int. Symp. Broadband Multimedia Syst. Broadcast. (BMSB), pp. 1-6, Jun. 2017.

Authorized licensed use limited to: University of Missouri Libraries. Downloaded on March 03,2026 at 23:52:00 UTC from IEEE Xplore. Restrictions apply.